



COM3502-4502-6502 SPEECH PROCESSING

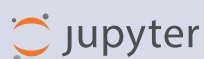
Lecture 1

Introduction



The Course

- Speech
 - speaking and hearing
 - acoustics and sound
 - the nature of speech
 - sounds and symbols
 - phonetics
 - phonology
 - prosody
- Speech Processing
 - signals and spectra
 - sampling
 - waveform processing
 - the Fourier transform
 - filters and linear prediction
 - cepstral analysis



Recommended Text Books



Teaching Staff



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Lecturer



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Graduate
Teaching
Assistants



Robbie Sutherland



Jason Clarke





Lecture material on Blackboard:
<https://vle.shef.ac.uk>

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Logistics

Module Map (Timing):

The lectures will take place Mondays 10 - 11 am in the [Diamond](#), [Lecture Theatre 5](#) and Tuesdays 9 - 10 am in the [Diamond](#), in [Lecture Theatre 4](#)

The computer labs will be in the [Diamond](#) in [Computer Rooms 1 and 2](#), [Thursdays](#) at 12 - 1 pm

This information should also be available in your personal [Time Table](#)

The following table details when which lectures will take place:

Lecture topic	Lecture date	Lab date	Lab Topic
L1 - Introductions	week 1 (Monday, 25. Sep. 2023, 10-11 am and Tuesday, 26. Sep. 2023, 9-10 am)	week 1 no lab	
L2 - Sound	week 2 (Monday, 02. Oct. 2023, 10-11 am and Tuesday, 03. Oct. 2023, 9-10 am)	week 2 (Thursday, 5 Oct. 2023, 12-1 pm)	Lab Sheet 1 - Signals
L3 - Signals	week 3 (Monday, 09. Oct. 2023, 10-11 am and Tuesday, 10. Oct. 2023, 9-10 am)	week 3 (Thursday, 12 Oct. 2023, 12-1 pm)	Lab Sheet 2
L4 - Digital Signals	week 4 (Monday, 16. Oct. 2023, 10-11 am and Tuesday, 17. Oct. 2023, 9-10 am)	week 4 (Thursday, 19 Oct. 2023, 12-1 pm)	Lab Sheet 3
L5 - Waveform Processing	week 5 NO LECTURE (reading week)	week 5 Reading week (no lab)	Reading week (no lab)
L6 - Frequency Analysis	week 6 (Monday, 30. Oct. 2023, 10-11 am and Tuesday, 31. Oct. 2023, 9-10 am)	week 6 (Thursday, 02 Nov. 2023, 12-1 pm)	Lab Sheet 4
L7 - The Z Transform	week 7 (Monday, 06. Nov. 2023, 10-11 am and Tuesday, 07. Nov. 2023, 9-10 am)	week 7 (Thursday, 09 Nov. 2023, 12-1 pm) (Assignment Handout, 9 Nov. 2023)	Lab Sheet 5
L8 - The Fourier Transform	week 8 (Monday, 13. Nov. 2023, 10-11 am and Tuesday, 14. Nov. 2023, 9-10 am)	week 8 (Friday, 16. Nov. 2023, 12-1 pm)	Lab Sheet 6
Reading week (no lecture)	week 9 (Monday, 20. Nov. 2023, 10-11 am and Tuesday, 21. Nov. 2023, 9-10 am)	week 9 (Friday, 23 Nov. 2023, 12-1 pm)	Lab Sheet 7
L9 - Linear Filters	week 10 (Monday, 27. Nov. 2023, 10-11 am and Tuesday, 28. Nov. 2023, 9-10 am)	week 10 (Friday, 30 Nov. 2023, 12-1 pm)	Lab Sheet 8
L10 - Speaking	week 11 (Monday, 04 Dec. 2023, 10-11 am and Tuesday, 05. Dec. 2023, 9-10 am)	week 11 no lab Assignment Submission due: 15 Dec. 2023)	
L11 - Hearing			
L12 - The Nature of Speech			
L13 - Sounds and Symbols			
L14 - Articulatory Phonetics			
L15 - Acoustic Phonetics			
L16 - Phonology and Prosody			
L17 - Adaptive Filtering			
L18 - Linear Prediction			
L19 - Cepstral Analysis			
L20 - Final Overview			



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- Lectures
 - 20x 50 mins (*with breaks*)
 - 2 per week
- Practical work
 - Weekly lab sheets (first in week 2)
 - Main programming assignment (~9 weeks work)
- Feedback/Interaction with Teaching Staff
 - Ask questions during lecture! Beneficial for everyone
 - Blackboard Discussion Group (*lectures + practical work*)
 - Sample Solutions for Lab Sheets
 - Staff contact details on Blackboard (<https://vle.shef.ac.uk>)
- Assessment
 - Main programming assignment (*worth 55%*)
 - Blackboard exam (*worth 45%*)
- Lecture notes (*these slides*)
 - available on Blackboard prior to each lecture

What is Speech Processing ?



“... the study of **speech signals** and the processing methods of these signals”

“... a special case of **digital signal processing** applied to speech signals”

http://en.wikipedia.org/wiki/Speech_processing



WIKIPEDIA
The Free Encyclopedia

What is Speech Processing For ?



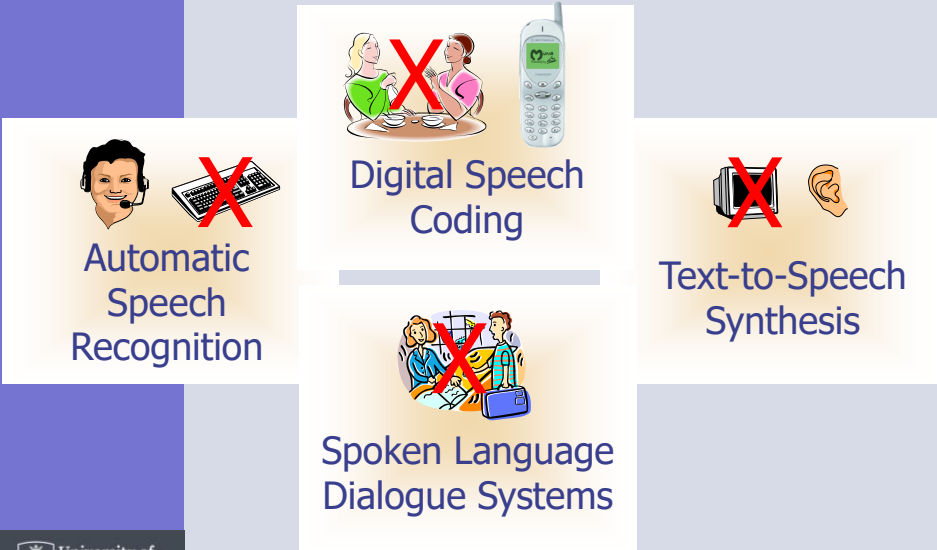
**THERE ARE OFFICIALLY MORE MOBILE DEVICES
THAN PEOPLE IN THE WORLD**

<https://www.independent.co.uk/life-style/gadgets-and-tech/news/there-are-officially-more-mobile-devices-than-people-in-the-world-9780518.html>

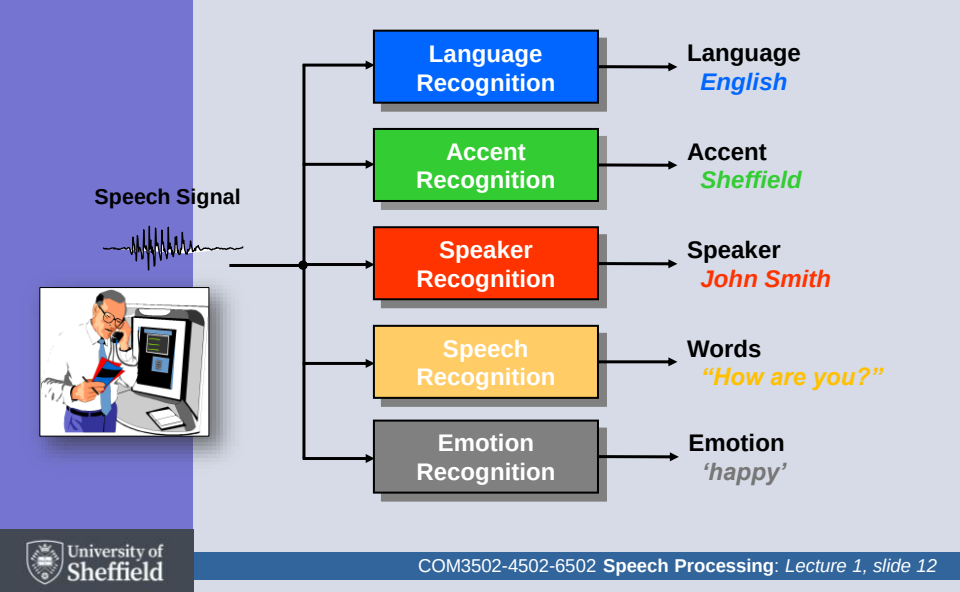
1876: **2**

2020: **9,200,000,000**

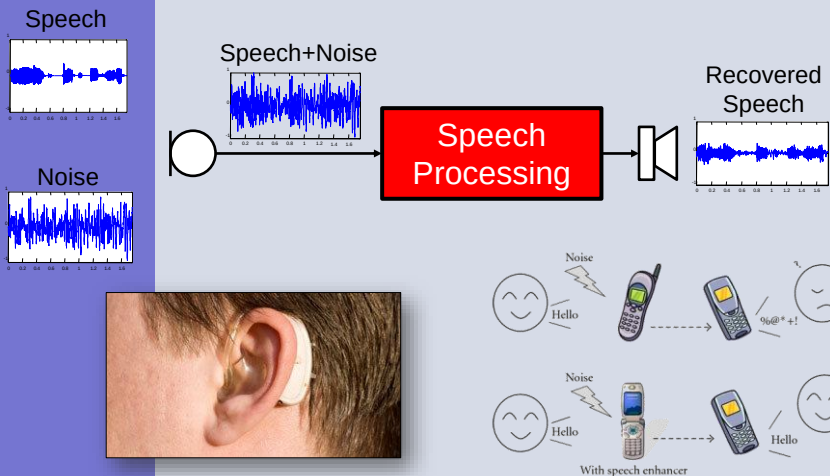
Speech Processing Technologies



Extracting Information from Speech



Speech Enhancement



Significant Market Penetration



Why Speech Processing ?

Lots of applications

... especially in Science Fiction !



[Star Trek IV: The Voyage Home \(1986\)](#)

Why use Speech ?

*“Speech is the ‘natural’ way to
interact with your computer”*

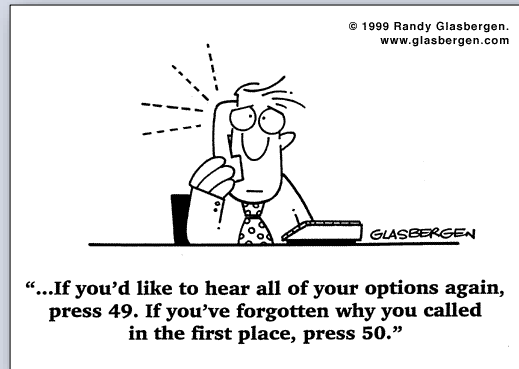
Wrong!

Speech *may* be a more intuitive way of
accessing information, controlling
things and communicating ...

but there may be viable alternatives

Why use Speech ?

Some alternatives can be problematic...



The Advantages of Speech



- hands-free
- eyes-free
- fast
- intuitive



Fran Capo

World Record Holder

603.32 wpm



*"You have been learning
skill needed to operate"*



Command & Control: *Vehicle*



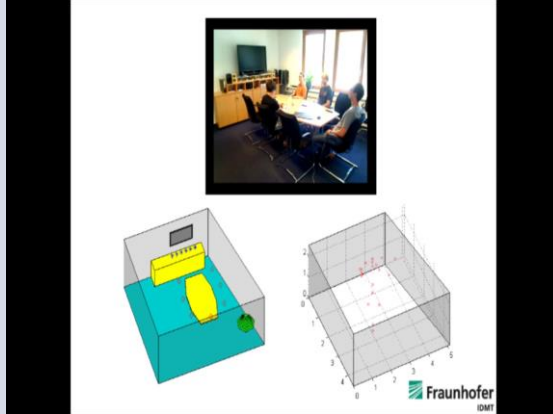
Robust Speaker-Independent Small-Vocabulary Automatic Speech Recognition

Intelligence: *Voice Stress Analysis*



Vocal Emotion Detection

Processing: *Speaker Localisation*



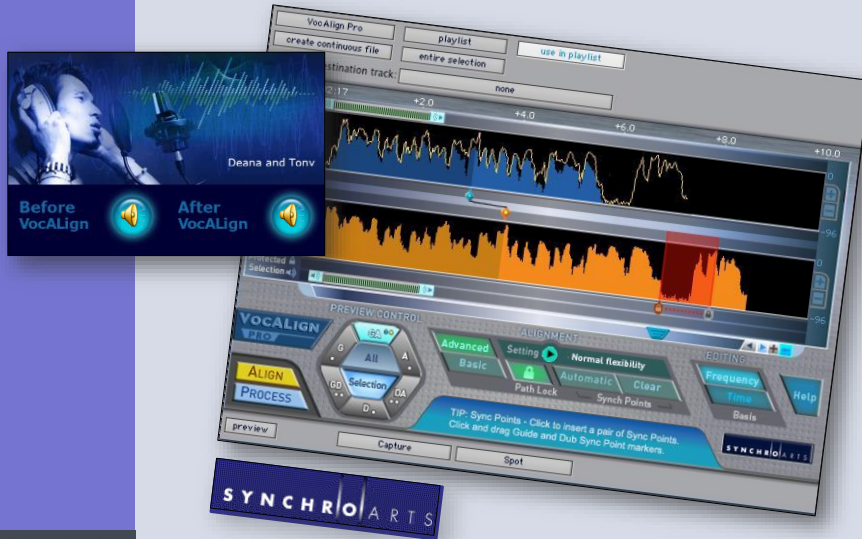
Direction/Position Estimation

Processing: *Special Effects*



Vocal Manipulation

Processing: *Audio Alignment*

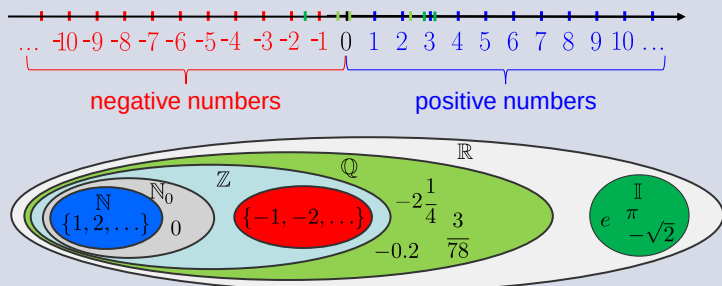


39

(Some) Notation & Basics

40

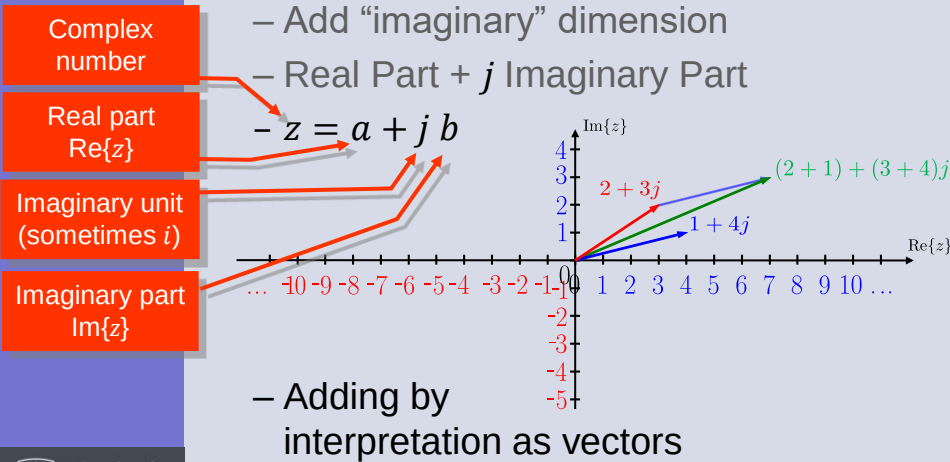
Repetition: Real Numbers



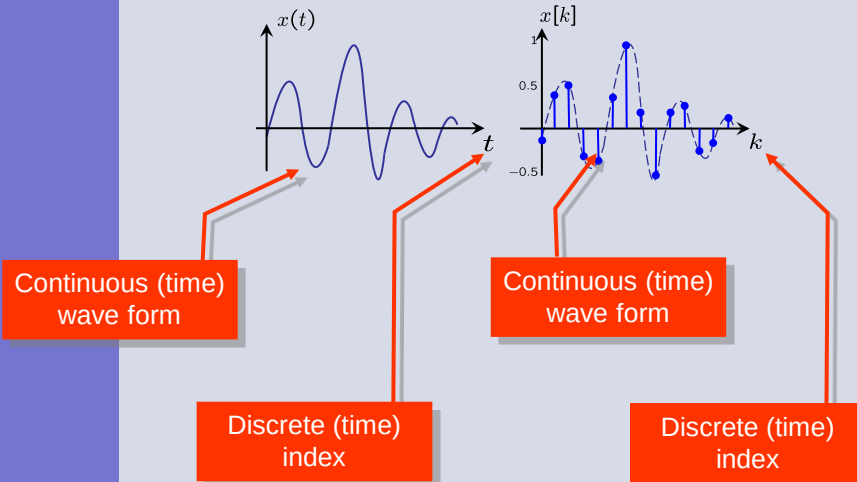
- Natural numbers $\mathbb{N} = \{1, 2, 3, \dots\}$
- Whole numbers $\mathbb{N}_0 = \{0, 1, 2, 3, \dots\}$
- Integer numbers $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, 3, \dots\}$
- Rational numbers $\mathbb{Q} = \{a/b : a \in \mathbb{Z} \text{ and } b \in \mathbb{N}\}$
every number resulting from a ratio
- Real numbers $\mathbb{R}$ (includes e.g. $\pi = 3.14159\dots$)

Complex Numbers

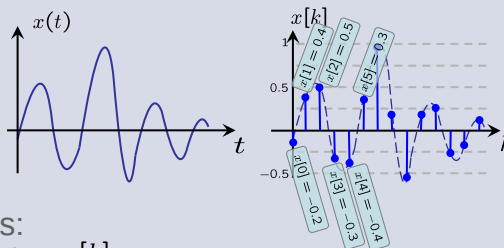
- Complex numbers $\mathbb{C}$
 - Add “imaginary” dimension
 - Real Part + j Imaginary Part



Continuous / Discrete Time



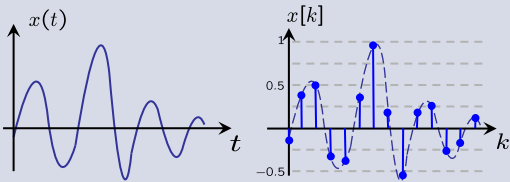
Vector Notation



- Scalars:
 - Signals: $x[k]$
 - Vectors:
 - Signal vectors: $\mathbf{x}[k] = [x[k], x[k-1], \dots, x[k-K+1]]^T$
- $[-0.2, 0.4, 0.5, -0.3, -0.4, 0.3, \dots]^T$

K : length of sequence

Notation



- Scalars:
 - Signals: $x[k]$
- Vectors:
 - Signal vectors: $\mathbf{x}[k] = [x[k], x[k-1], \dots, x[k-K+1]]^T$
- Matrixes:
$$\mathbf{A}[k] = \begin{bmatrix} a_{00}[k] & a_{01}[k] & \cdots & a_{0K}[k] \\ a_{10}[k] & a_{11}[k] & \cdots & a_{1K}[k] \\ \vdots & \vdots & & \vdots \\ a_{M0}[k] & a_{M1}[k] & \cdots & a_{MK}[k] \end{bmatrix}$$

Notation

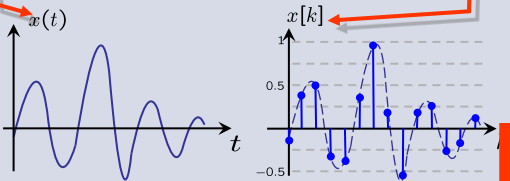
(time) continuous signal

discrete signal

'normal' letters

bold letters

bold capital letters



(squared) brackets

	Continuous time domain	Discrete time domain	Continuous freq. domain	Discrete freq. domain
Scalar	$x(t), \xi(t)$	$x[k], \xi[k]$	$x(e^{j\Omega}), \xi(e^{j\Omega})$	$x[n], \xi[n]$
Vector	$\mathbf{x}(t), \boldsymbol{\xi}(t)$	$\mathbf{x}[k], \boldsymbol{\xi}[k]$	$\mathbf{x}(e^{j\Omega}), \boldsymbol{\xi}(e^{j\Omega})$	$\mathbf{x}[n], \boldsymbol{\xi}[n]$
Matrix	$\mathbf{X}(t), \boldsymbol{\Xi}(t)$	$\mathbf{X}[k], \boldsymbol{\Xi}[k]$	$\mathbf{X}(e^{j\Omega}), \boldsymbol{\Xi}(e^{j\Omega})$	$\mathbf{X}[n], \boldsymbol{\Xi}[n]$

(round) parentheses

(squared) brackets

(round) parentheses

Notation – Part 1

- Scalars: $x[k]$
 - Discrete Time index
 - Impulse responses (time-varying): $h_i[k]$
 - Coefficient index
 - Example for a (real) convolution: $y[k] = \sum_{i=0}^{K-1} x[k-i]h_i[k]$
- Vectors:
 - Signal vectors: $\mathbf{x}[k] = [x[k], x[k-1], \dots, x[k-K+1]]^T$
 - Impulse response vectors (time-varying): $\mathbf{h}[k] = [h_0[k], h_1[k], \dots, h_{K-1}[k]]^T$
 - Example for a (real) convolution: $y[k] = \mathbf{x}^T[k]\mathbf{h}[k] = \mathbf{h}^T[k]\mathbf{x}[k]$
- Matrixes:
$$\mathbf{A}[k] = \begin{bmatrix} a_{00}[k] & a_{01}[k] & \cdots & a_{0K}[k] \\ a_{10}[k] & a_{11}[k] & \cdots & a_{1K}[k] \\ \vdots & \vdots & \ddots & \vdots \\ a_{M0}[k] & a_{M1}[k] & \cdots & a_{MK}[k] \end{bmatrix}$$

Notation – Part 2

- Auto and cross correlation with real, stationary stochastic processes:
 - Autocorrelation function: $E\{x[k]x[k+\kappa]\} = r_{xx}[\kappa]$
 - Cross correlation function: $E\{x[k]y[k+\kappa]\} = r_{xy}[\kappa]$
 - Auto-power density spectrum:
$$\phi_{xx}(\Omega) = \sum_{\kappa=-\infty}^{+\infty} E\{x[k]x[k+\kappa]\}e^{-j\Omega\kappa} = \sum_{\kappa=-\infty}^{+\infty} r_{xx}[\kappa]e^{-j\Omega\kappa}$$
 - Cross-power density spectrum:
$$\phi_{xy}(\Omega) = \sum_{\kappa=-\infty}^{+\infty} E\{x[k]y[k+\kappa]\}e^{-j\Omega\kappa} = \sum_{\kappa=-\infty}^{+\infty} r_{xy}[\kappa]e^{-j\Omega\kappa}$$
 - Continuous frequency index

This lecture has covered ...

- What speech processing is
- What speech processing is for
- Speech processing technologies
- The advantages of speech
- Types of application
- Some math notation

Any Questions ?

*Ask during lecture (e.g., now),
or post in the Blackboard Discussion group*



Next lecture ...

Sound